

RoboMind AI Challenge – 2026

Training Explanation (TEC05) – SAATHI

1. Phase I: Voice Interaction and Activation Training

The model is trained to operate in a standby state until a specific acoustic trigger is detected.

- **Wake-Word Detection:** The system is integrated with the **Speech to Text** engine to recognize the wake-word "**Scratch**".
- **Interactive Prompting:** Upon activation, the model is trained to issue a standardized vocal prompt: "**Tell me your symptoms**".
- **Input Recognition:** The model is trained to parse natural language phrases such as "**I am having...**" or "**I am suffering with...**" to extract key health indicators.

2. Phase II: Symptom Classification and Diagnostic Mapping

The core training involves mapping specific symptoms to a "Possible Issue." Based on the source data, the model has been trained on the following mappings:

- **Symptom: Headache** → **Possible Issue:** Tension headache or dehydration.
- **Symptom: Autoimmune Vasculitis** → **Possible Issue:** Autoimmune vasculitis.
- **Symptom: Chest Pain** → **Possible Issue:** Heart-related issue or anxiety.

3. Phase III: Quad-Layer Output Generation Training

For every identified condition, the model is trained to retrieve and deliver a structured four-part response:

A. Nutritional Protocol ("What to Eat" vs. "What to Avoid")

The data training links specific conditions to dietary interventions:

- **Headache:** Recommend hydrating foods (watermelon, cucumber); restrict caffeine and alcohol.
- **Vasculitis:** Recommend anti-inflammatory foods (fruits, vegetables, grains, fish); restrict processed foods, added sugars, and salt.
- **Chest Pain:** Recommend antioxidant-rich foods (berries, vegetables, oats); restrict saturated fats and sugars .

B. Specialist Referral Logic

The model associates conditions with the appropriate medical field:

- **Headache** → General practitioner.
- **Vasculitis** → Rheumatologist.
- **Chest Pain** → Cardiologist or primary care physician.

C. Emergency Safety Net (Red Flag Detection)

A critical training component is the identification of high-severity symptoms that trigger an **Emergency Warning**:

- **Headache Flags:** "Sudden and severe," "weakness," or "confusion".
- **Vasculitis Flags:** "Chest pain" or "difficulty breathing".
- **Chest Pain Flags:** "Severe," "persistent," "shortness of breath," or "dizziness".

4. Phase IV: Hardware Integration and Deployment

The integration logic ensures the model performs efficiently on edge computing hardware:

- **Environment:** The model is developed in **Python** and deployed on a **Raspberry Pi** using
- **Real-Time Processing:** Integration is optimized for low-latency voice-to-text and text-to-speech, allowing the bot to provide immediate feedback in a home setting.
- **Local Execution:** By running on the Raspberry Pi, the model is trained to handle the diagnostic logic locally, ensuring the health assistant remains functional and responsive